Unit 8, Lesson 5: Evaluating Numeric Expressions – Part 2

Benchmarks

MA.7.NSO.2.1 Solve mathematical problems using multi-step order of operations with rational numbers including grouping symbols, whole-number exponents, and absolute value.

MA.7.NSO.2.2 Add, subtract, multiply and divide rational numbers with procedural fluency.

Learning Target

- ☐ I can evaluate numeric expressions with up to three or four operations.

8.5.1 Warm-Up: Estimating Length

The roll of bubble wrap on the left is 1 foot high and 100 feet long.

Complete the following to guess the length of the bubble wrap on the right.

1. What is a guess that is too low?

2. Give a guess that does not make sense and explain how you know it doesn't make sense.

3. What is a guess that is too high?

4. What is your best guess of the length of the bubble wrap on the right when unrolled?

8.5.2 Guided Instruction: Using the Order of Operations on Expressions with Nested Grouping Symbols

The order of operations can be used to evaluate expressions using grouping symbols, rational numbers, exponents, and absolute values.

1. Two students evaluated the expression $\left(\frac{(3+0.5)}{0.5}\right)^3$. This expression shows nested grouping symbols, or groupings inside of other groupings.

Abby	Devon
$\left(\frac{(3+0.5)}{0.5}\right)^3$ $\left(\frac{3.5}{0.5}\right)^3$ $(7)^3$ 343	$\left(\frac{(3+0.5)}{0.5}\right)^3$ $\frac{(3+0.5)^3}{0.5^3}$ $\frac{3.5^3}{0.5^3}$ $\frac{42.875}{0.125}$ 343

How were Abby's steps and Devon's steps different from each other?

When using the order of operations to evaluate an expression with nested grouping symbols, apply the order of operations inside the inner grouping symbols first.

2. Zion says since multiplication is commutative and the order does not matter, he thinks he can evaluate the expression $[0.05 \times (3 \times 2)^4]^2$ by either multiplying 0.05×3 first or by multiplying 3×2 first.

   Discuss with your partner if Zion is correct when he stated that both would work.

3. Telena says the problem can be worked either by following the order of operations or by using the laws of exponents. Complete the table to show the work for both possibilities to determine if Telena is correct.

Order of Operations	Laws of Exponents
$\left[0.1 \cdot \left(\frac{2}{3} \cdot -3\right)^3\right]^2$	$\left[0.1 \cdot \left(\frac{2}{3} \cdot -3\right)^3\right]^2$

8.5.3 Exploration: Grouping Symbols

1. Without working the problems, discuss with your partner which expression in the table you expect to have the largest value.

Expression A	Expression B
$10 - (0.75 \times 12) + 4^2$	$(10 - 0.75) \times (12 + 4^2)$
Expression C	**Expression D**
$10 - 0.75 \times 12 + 4^2$	$10 - 0.75 \times (12 + 4)^2$

2. Put an asterisk (*) next to the expression you expect will have the largest value. Write your explanation here.

3. Evaluate each of the expressions in the table with your partner.

4. With another set of partners, share if your guess was or was not correct. Discuss how the order of operations affected the value of your guesses. Summarize your discussion.

5. For the two expressions in the table that resulted in the same value, explain why they have the same value.

8.5.4 Collaboration: Using the Order of Operations to Evaluate Expressions with Nested Grouping Symbols

1. The expressions in each row have the same value. Evaluate the expressions in one column while your partner evaluates the expressions in the other column. Compare values and if they are not equivalent, work together to find and correct the error(s).

Partner A	Value of Both Expressions	Partner B
$\lvert(35.5 - 5^2) \div 3\rvert$		$\lvert 4.3 \times 5 - 5^2 \rvert$
$32 \div [(-2)^2 \times 5 - 4]$		$[(-2)^3 + 16] \div (-2)^2$
$[(18 - 4 \times 5) \times 3]^4$		$[(2.4 - 5.4) \times (-2)^1]^4$
$11 + (0.3^0 + 3)^2$		$[36 \div (2^3 + 4)]^3$

8.5.5 Your Turn!

1. Evaluate each expression without using a calculator.

   a. $[(10-15)\times 2]^3+50$

   b. $(2^4-3\times 6)^2$

   c. $|18+(-2)\times 14|^5$

   d. $[0.8\times(1.2+2.8)^2]^2$

   e. $[(0.8\times 10)^2+4.5]\div 0.5$

   f. $(10\times 0.4)^2+\big(28.4\div(-4)\big)$

8.5.6 Wrap-Up: Reflection

1. Complete each statement based on your learning today.

    - A barrier to my learning was …

    - I will try to overcome this by …

    - Today I am still unsure about …

    - To fill in this gap I intend to …